

A Curvature-Variable Propagation Framework for Effective Causal Deviation in Relativistic and Quantum-Geometric Systems

Introducing the Dillon Equation as a Falsifiable Deviation Observable

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Abstract

This paper proposes a curvature-variable propagation framework in which locally invariant causal propagation is preserved while effective propagation across extended nonuniform geometry is represented by a curvature-dependent quantity, denoted c_{eff} . The central relation, referred to here as the Dillon Equation, is written in compact form as

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

where K denotes curvature state or curvature topology, R denotes a geometric curvature contribution, ∇K denotes curvature-gradient structure, and $\partial_t K$ denotes temporal curvature evolution. The associated deviation observable is

$$\Delta c_{\text{eff}} = c_{\text{eff}} - c = f(K, R, \nabla K, \partial_t K) - c.$$

The framework does not assert that the locally measured speed of light varies. Instead, it distinguishes local Lorentz-invariant causal structure from effective propagation across structured physical domains. In the limiting case where curvature-gradient and temporal-curvature contributions vanish, $c_{\text{eff}} \rightarrow c$ and $\Delta c_{\text{eff}} \rightarrow 0$, recovering the Einsteinian local limit. A normalized first-order model is introduced to constrain the otherwise general functional form and to generate falsifiable signatures in timing residuals, lensing residuals, electromagnetic phase measurements, quantum coherence experiments, and cosmological structure observations. The paper is presented as a candidate theoretical framework rather than a completed theory of quantum gravity or a replacement for the Standard Model. Related Dillon Architecture, Curvature Variable Physics, Omega-Genesis, and Omega Calculus manuscripts are cited as programmatic context and as a public review archive, while the present paper is constructed to stand independently on its propagation claim, limiting conditions, falsifiability criteria, dimensional normalization, and minimum empirical test.

Keywords: effective propagation; curvature-variable geometry; causal deviation; Lorentz invariance; quantum geometry; gravitational lensing; cosmology; falsifiability; Dillon Equation

1 Introduction

The problem of unification in physics is not merely the problem of combining mathematical formalisms. It is the problem of identifying the common physical object through which gravitation, field propagation, quantum state behavior, measurement, energy transfer, and cosmological evolution may be compared without discarding the experimentally successful limits of existing theories.

General relativity describes gravitation as spacetime geometry [3, 4, 13, 12]. Quantum mechanics and quantum field theory describe matter and interaction in terms of states, amplitudes, operators, and fields [5, 7, 15]. Maxwell’s theory unifies electricity, magnetism, and light as field propagation [1]. Black-hole thermodynamics and quantum-field effects in curved spacetime suggest that geometry, information, entropy, and causal horizons are deeply linked [8, 9, 14].

The present manuscript proposes that effective propagation may serve as a unifying comparative object. The locally measured speed of light, denoted c , remains invariant in accordance with special relativity and local Lorentz invariance. However, propagation across extended, structured, nonuniform, or dynamically evolving geometry may be represented by an effective propagation term c_{eff} . The Dillon Equation expresses this effective propagation as a function of curvature state, geometric curvature, curvature-gradient structure, and temporal curvature evolution.

The purpose of this paper is not to claim completion of a final theory of everything. Its purpose is narrower and more testable: to define a curvature-variable propagation framework, identify its limiting behavior, introduce a deviation observable, and specify conditions under which the framework could be supported or falsified.

2 Scope and Status of the Claim

Scope and status. The Dillon Equation is presented as a candidate curvature-variable propagation framework. It is not claimed here as an experimentally confirmed replacement for general relativity, quantum field theory, or the Standard Model. Its purpose is to define a falsifiable deviation observable, Δc_{eff} , and to identify limiting cases, dimensional constraints, test domains, and mathematical conditions under which the framework may be evaluated.

The framework should therefore be read as a proposed research program with a specific test object: effective causal deviation across structured nonuniform geometry. The claim is intentionally narrower than a completed theory of everything. If the deviation observable cannot be made operational, or if the first-order coefficients are constrained to zero in the relevant domains, then the proposed first-order implementation fails in those domains.

3 Relationship to the Dillon Architecture and Prior CVP Work

The Dillon Equation is introduced here as the propagation-focused formalization of a broader research program referred to as the Dillon Architecture. In that research stack, Curvature Variable Physics (CVP) functions as the field-geometry language; the Dillon Equation defines an effective propagation relation; and the deviation observable Δc_{eff} supplies the proposed test target. The broader Dillon Architecture has been described as a manuscript stack spanning CVP, Omega-Genesis, fusion, cryptography, deterministic systems, quantum gravity and cosmology, aerospace and electromagnetic modeling, and bounded autonomous control [21].

This relationship is intentionally narrow. The present paper does not require acceptance of every applied branch of the Dillon Architecture. Instead, it isolates one journal-evaluable claim: effective propagation across structured geometry may be represented by a curvature-variable function while preserving local invariant causality as the Einstein limit. Prior Dillon Architecture materials motivate the notation and programmatic lineage, but they are not treated as experimental confirmation of the present propagation framework.

Earlier CVP work has applied a curvature-response operator to the interpretation of Higgs-sector mass stabilization while preserving the Standard Model Higgs mechanism as the experimentally

established scalar-field baseline [22]. In that prior formulation, the curvature-response operator is written schematically as

$$C(K) = \beta_1 \partial_t K + \beta_2 \nabla K + \beta_3 \nabla^2 K,$$

and the proposed effective mass relation is expressed as

$$m_{\text{eff}} = m_0 + \lambda_H H + \lambda_C C(K).$$

The present paper generalizes the same response logic from mass stabilization to effective causal propagation. The relevant shared concept is not a claim that Standard Model physics is discarded; it is the proposal that stable physical behavior may be interpreted through curvature-mediated constraint layers.

Applied Dillon Architecture materials also extend the curvature-state grammar into electromagnetic propagation, energy harvesting, aerospace, and plasma-control settings [23]. Those applications are not used here as proof of the Dillon Equation. They are cited as candidate engineering domains in which a curvature-state vector, response operator, safety projection, evidence logging, and falsifiable boundary accounting could be tested. For a journal treatment, the important separation is this: the theoretical claim is Δc_{eff} ; the applied branches are future validation environments.

4 Relationship to 4D Predictive Geometry

The present manuscript is also related to prior Dillon Framework work on 4D predictive geometry, vortex geodesic topology, and toroidal embedding [24, 25]. That work treats cyclic and phase-coupled systems as better represented on compact manifolds than on flat Euclidean feature spaces. In the 4D Innovation white paper, system state is embedded on a toroidal manifold parameterized by radii (R, r) and angular coordinates (Θ, Φ) , with an explicit phase/time coordinate. The same paper introduces harmonic control planes indexed by $\{3, 6, 9\}$, geodesic modulation through an action objective, equilibrium-node damping, targeted outcome alignment, and a measurement-first validation program.

For the purposes of the present journal paper, the 4D and predictive-geometry materials are not treated as proof of the Dillon Equation. They are cited as methodological background and as possible implementation scaffolding for simulation, reachability analysis, and reproducible testing of curvature-variable propagation hypotheses. In particular, a toroidal or compact-manifold representation may be useful when the relevant observable is phase, periodicity, wrap-around continuity, or trajectory stability under curvature-sensitive constraints.

The relationship can be summarized as follows. The Dillon Equation proposes the physical deviation observable Δc_{eff} . Predictive Geometry supplies a forward-state language for reachability corridors, instability corridors, attractor forecasts, and rigidity thresholds. The 4D toroidal framework supplies a candidate computational geometry for bounded phase-coupled evolution. The present manuscript isolates the narrower physical claim: effective propagation may admit curvature-variable deviation while preserving the Einstein local limit.

A journal-safe interpretation is therefore that 4D Predictive Geometry is a modeling environment, not an empirical confirmation. It may help generate synthetic test cases, phase-residual simulations, manifold-constrained trajectories, ablation studies, and reproducible null tests. If later work demonstrates that toroidal/geodesic simulation improves the prediction of propagation residuals under independently measured curvature conditions, that would support the computational usefulness of the 4D layer. It would not, by itself, prove the physical theory without external measurement and replication.

5 Related Dillon Framework Manuscripts and Citation Discipline

The related-manuscript citation approach used here is conservative. A Dillon Framework manuscript is cited in the main body only when it directly supports one of the following elements: the canonical distinction between local invariant c and geometry-conditioned c_{eff} ; the curvature-response operator; the deviation-observable test program; a concrete candidate validation domain; or the methodological reduction grammar used to organize the framework.

Three additional Dillon Framework sources are therefore cited in the main body. First, the Predictive Quantum Measurement Overlay with 4D Adaptive Electromagnetic Propagation Control manuscript is relevant because it states the canonical Dillon Equation standard, distinguishes local invariant causal speed from geometry-conditioned effective propagation, and frames 4D Predictive Geometry as a reachability-corridor layer for quantum-memory-enhanced measurement and adaptive electromagnetic control [26]. Second, the CVP Artemis Lunar Descent manuscript is relevant because it applies the same propagation expression and curvature-response operator to a narrow aerospace problem: mascon-rich terminal lunar descent and bounded convergence under dynamic curvature modeling [33]. Third, the broad architecture paper, *From Collatz to Cure*, is relevant because it states the canonical equation family while explicitly characterizing broader mathematical, cryptographic, and physical claims as requiring independent specialist review, replication, formal verification, and domain-specific validation [28].

Other Dillon Framework items are retained in Appendix D as contextual related work rather than as direct support for the central physics claim. This distinction is important for peer review. The present paper does not require acceptance of broad claims in cryptography, energy, defense, or mathematical closure. It isolates the narrower proposition that effective propagation may admit curvature-variable deviation while preserving the local Einstein limit.

6 Relationship to Omega-Genesis, Omega Calculus, and Public Reproducibility

The Dillon Equation is also related to the author’s Omega-Genesis and Omega Calculus manuscripts. In the author’s stated architecture, Omega-Genesis supplies a mathematical reduction-and-closure layer, Omega Calculus supplies the symbolic language for closure and admissibility, the Dillon Equation supplies the curvature-variable propagation layer, and applied systems such as CVP Spin Overlay and TEVR supply implementation and evidence layers. This relationship is programmatic and methodological; it is not used as a premise in the physical argument of the present paper.

The Omega Calculus compact companion manuscript defines Omega Calculus as a language layer for convergence, closure, curvature-governed evolution, recursive reduction, leak detection, rigidity response, and attractor structure across mathematical, physical, and engineered systems [30]. Its definitions of Omega states, Omega flow, Omega closure, Omega leaks, Omega attractors, and Omega rigidity provide useful terminology for describing structural determinism without collapsing separate domains into a single unverified theorem.

The author also maintains a public Omega-Genesis repository containing manuscripts, code, simulations, and validation artifacts for independent review. The repository describes OmegaGenesis as a structural framework for analyzing the Collatz system through a finite modular architecture with a four-family dynamical partition, finite residue-graph core, and invariants for drift and valuation [32]. The author has stated these materials for peer review and independent verification.

Related proof manuscripts, including the author’s P versus NP manuscript, are cited only as companion mathematical-closure work. The P versus NP manuscript asserts a structural proof

through differentiable information manifolds, modular Ricci flow, residual families, and a finite near-critical endgame closure chain [31]. Those claims are not assumed by the present physics manuscript. The physical claim here remains limited to the curvature-variable propagation framework, the Einstein limiting condition, the normalized deviation observable, and the proposed falsifiability program. This separation is intentional: the Dillon Equation paper should be reviewable even by readers who reserve judgment on the broader Omega-Genesis proof corpus.

7 Historical and Theoretical Motivation

The sequence of physical unification has repeatedly transformed the meaning of propagation. Newtonian mechanics treated action through force and acceleration. Maxwell showed that light is an electromagnetic wave and that field propagation has a characteristic finite speed [1]. Einstein made causal structure central by establishing the invariant speed of light in special relativity and then reinterpreting gravitation as spacetime geometry in general relativity [2, 3]. Quantum theory introduced wavefunctions, operators, uncertainty, quantization, and the problem of measurement [5, 6, 7]. Later developments tied gravity to thermodynamics, horizons, and entropy [8, 9].

These historical developments suggest a recurring pattern: the most successful unifications identify a deeper invariant or structural relation behind apparently separate phenomena. The framework developed here follows that pattern by asking whether effective propagation through curvature-variable geometry can be formalized as a common descriptor across domains.

8 Assumptions and Non-Claims

To prevent misinterpretation, the framework is stated with explicit assumptions and non-claims.

8.1 Assumptions

Postulate 1 (Local causal invariance). *In a sufficiently local inertial frame, causal propagation is governed by the invariant speed c . The framework preserves local Lorentz invariance as a required limiting condition.*

Postulate 2 (Effective extended propagation). *Across an extended physical region with nonuniform or dynamically evolving geometry, the measured or inferred propagation behavior of a process may be represented by an effective propagation quantity c_{eff} .*

Postulate 3 (Curvature-gradient contribution). *Spatial variation in curvature may contribute to the difference between c_{eff} and the local invariant limit c .*

Postulate 4 (Temporal-curvature contribution). *Time-dependent curvature evolution may contribute to the difference between c_{eff} and the local invariant limit c .*

8.2 Non-Claims

This paper does not claim that the locally measured speed of light is variable. It does not reject special relativity or general relativity. It does not provide a completed theory of quantum gravity. It does not replace the Standard Model of particle physics. It does not assert that a single untested scalar function is sufficient to explain all known physical phenomena. Rather, it proposes a constrained propagation framework whose central observable, Δc_{eff} , may be tested against timing, phase, lensing, coherence, and cosmological residuals.

9 Notation

Table 1: Mathematical notation used in the Dillon Equation framework.

Symbol	Meaning
c	Locally invariant causal speed; the Einsteinian local limit.
c_{eff}	Effective propagation across an extended, structured, or nonuniform physical region.
Δc_{eff}	Effective propagation deviation, defined as $c_{\text{eff}} - c$.
K	Curvature state, curvature topology, or generalized curvature structure.
R	Geometric curvature contribution; may correspond to scalar curvature or a Ricci-type contraction in a specific model.
∇K	Spatial curvature-gradient structure.
$\partial_t K$	Temporal curvature-evolution term.
K_0, R_0, G_0, T_0	Reference scales used to normalize curvature, geometric curvature, curvature gradient, and temporal curvature evolution.
F	Dimensionless normalized propagation function.
$\alpha, \beta, \gamma, \delta$	Coupling coefficients to be constrained experimentally or observationally.
ϵ	Dimensionless fractional deviation, $\epsilon = \Delta c_{\text{eff}}/c$.

10 Curvature-Variable Propagation Framework

Definition 1 (Dillon Equation). *The compact curvature-variable propagation relation is defined as*

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K). \quad (1)$$

Equation (1) is not yet a complete physical model. It is a framework equation identifying the variables on which effective propagation is hypothesized to depend. A completed domain-specific model must specify the manifold, metric structure, curvature definitions, reference scales, coupling coefficients, and observational map from c_{eff} to measurable quantities.

Definition 2 (Deviation observable). *The effective propagation deviation is defined as*

$$\Delta c_{\text{eff}} = c_{\text{eff}} - c = f(K, R, \nabla K, \partial_t K) - c. \quad (2)$$

The deviation observable is the central object of empirical interest. The model is not evaluated by asking whether c locally varies, but by asking whether an effective propagation residual remains after standard local and relativistic effects have been accounted for.

11 Einstein Limit and Local Invariance

Proposition 1 (Einstein local-limit recovery). *If curvature-gradient and temporal-curvature contributions vanish or become observationally negligible, and if the normalized propagation function satisfies the local-limit condition, then the Dillon Equation recovers local relativistic propagation:*

$$\nabla K \rightarrow 0, \quad \partial_t K \rightarrow 0 \quad \implies \quad c_{\text{eff}} \rightarrow c, \quad \Delta c_{\text{eff}} \rightarrow 0. \quad (3)$$

Proof sketch. By Postulate 1, the sufficiently local inertial-frame limit is governed by invariant causal speed c . If curvature-gradient and temporal-curvature terms vanish, no extended nonuniform correction remains in the proposed propagation function. Requiring consistency with special relativity imposes $f(K, R, 0, 0) \rightarrow c$ in the locally uniform limit. Substitution into Eq. (2) gives $\Delta c_{\text{eff}} \rightarrow 0$. $\square$

This proposition is not optional. It is a consistency requirement. Any implementation of the Dillon Equation that fails to recover the Einstein limit is outside the intended framework.

12 Normalized Functional Model

The compact form in Eq. (1) is too general for direct empirical assessment. A minimal normalized model is therefore introduced:

$$c_{\text{eff}} = c F\left(\frac{K}{K_0}, \frac{R}{R_0}, \frac{|\nabla K|}{G_0}, \frac{|\partial_t K|}{T_0}\right), \quad (4)$$

where F is dimensionless and satisfies

$$F(0, 0, 0, 0) = 1 \quad (5)$$

under a locally flat or reference-uniform limit. The corresponding fractional deviation is

$$\epsilon \equiv \frac{\Delta c_{\text{eff}}}{c} = F - 1. \quad (6)$$

12.1 Dimensional Normalization and Scale Definitions

A journal-evaluable implementation must be dimensionally disciplined. The normalized variables in Eq. (4) are dimensionless ratios, not raw additions of unlike physical quantities. The reference scales have the following role:

- K_0 is the reference curvature-state scale for the selected domain.
- R_0 is the reference geometric curvature scale, such as a scalar-curvature, Ricci-type, or model-specific curvature benchmark.
- G_0 is the reference curvature-gradient scale against which $|\nabla K|$ is measured.
- T_0 is the reference temporal curvature-variation scale against which $|\partial_t K|$ is measured.

The values of K_0 , R_0 , G_0 , and T_0 cannot be universal constants unless a deeper theory supplies them. In the present framework they are domain-specific normalization scales that must be stated before coefficient estimation. This requirement prevents the model from combining dimensionally incompatible quantities and makes the coefficient vector $(\alpha, \beta, \gamma, \delta)$ interpretable.

12.2 First-Order Approximation

For sufficiently small normalized deviations, a first-order expansion may be written as

$$\frac{\Delta c_{\text{eff}}}{c} \approx \alpha \frac{K}{K_0} + \beta \frac{R}{R_0} + \gamma \frac{|\nabla K|}{G_0} + \delta \frac{|\partial_t K|}{T_0}. \quad (7)$$

The coefficients $\alpha, \beta, \gamma, \delta$ are not assumed. They must be estimated, bounded, or ruled out by observation or experiment.

Proposition 2 (Falsifiable coefficient constraint). *If all admissible experiments and observations impose $\alpha = \beta = \gamma = \delta = 0$ within experimental uncertainty over the relevant domain, then the first-order Dillon deviation model is unsupported in that domain.*

Proof sketch. Equation (7) defines the first-order deviation as a linear combination of normalized curvature variables. If each coefficient is experimentally constrained to zero within uncertainty, the model predicts no first-order deviation. In that domain, the first-order implementation does not add predictive content beyond the local relativistic limit. $\square$

13 Theory-of-Everything Criteria

A candidate unification framework should satisfy at least three minimal criteria:

Criterion 1. Continuity. It must recover successful existing theories as limiting cases.

Criterion 2. Unification. It must identify a common mathematical or physical object by which distinct domains can be compared.

Criterion 3. Falsifiability. It must produce measurable predictions that could fail.

The Dillon Equation is proposed as satisfying these criteria as follows. Continuity is enforced through the Einstein limit. Unification is attempted by treating effective propagation as a common object across gravitational, electromagnetic, quantum-informational, and cosmological settings. Falsifiability is introduced through Δc_{eff} and the coefficient constraints in Eq. (7).

14 Propositions

Proposition 3 (Deviation is not local light-speed violation). *A nonzero Δc_{eff} does not, by itself, imply violation of local Lorentz invariance.*

Proof sketch. The deviation is defined between an effective extended propagation quantity and the local invariant limit. Local Lorentz invariance concerns sufficiently local inertial measurements. A propagation residual inferred across an extended structured region may be nonzero while local measurements remain invariant. The two claims concern different operational regimes. $\square$

Proposition 4 (Curvature-gradient sensitivity). *If $\gamma \neq 0$ in Eq. (7), then domains with larger normalized curvature gradients should exhibit larger fractional propagation residuals, all else being equal.*

Proof sketch. Holding the remaining normalized variables fixed, Eq. (7) is monotonic in $|\nabla K|/G_0$ when γ has fixed nonzero sign. Thus a systematic residual correlated with curvature-gradient magnitude is predicted. $\square$

Proposition 5 (Temporal-curvature sensitivity). *If $\delta \neq 0$ in Eq. (7), then time-varying curvature environments should exhibit propagation residuals correlated with $|\partial_t K|/T_0$.*

Proof sketch. The temporal-curvature term contributes linearly to the first-order fractional deviation. If δ is nonzero, changes in normalized temporal curvature evolution must alter $\Delta c_{\text{eff}}/c$ after other terms are controlled. $\square$

15 Physical Interpretation

The framework can be summarized as follows: locally, causality is governed by c ; across extended structured geometry, propagation may acquire effective behavior governed by curvature variables. This is analogous in spirit to distinguishing a local invariant law from its expression through a medium, geometry, or boundary condition, while avoiding the claim that spacetime is a conventional material medium.

Potential observational expressions of Δc_{eff} include timing residuals, phase offsets, coherence shifts, lensing residuals, redshift residuals, and horizon-scale propagation signatures. The framework becomes physically meaningful only when such residuals are mapped to explicit observables and compared against existing models.

16 Falsifiability Criteria

The following criteria define failure modes for the framework. A model that cannot fail is not a physical theory.

Falsifiability Criterion 1 (Local-limit failure). *If a proposed Dillon model predicts nonzero local deviation in regimes where special relativity is experimentally confirmed, that model is rejected.*

Falsifiability Criterion 2 (Residual non-correlation). *If measured propagation residuals show no statistically significant correlation with any proposed normalized curvature variable after standard effects and systematics are controlled, the corresponding coefficient model is unsupported.*

Falsifiability Criterion 3 (Standard-model sufficiency). *If general relativity, quantum field theory, Maxwellian electrodynamics, and standard cosmological modeling fully account for the relevant residuals within uncertainty, no Dillon correction is warranted in that domain.*

Falsifiability Criterion 4 (Non-transferability). *If coefficients inferred in one domain fail to predict or bound behavior in a theoretically linked domain, the proposed universal form of the function F must be revised or rejected.*

17 Minimum Empirical Test

A minimum empirical test should be designed before broader claims are evaluated. The most direct near-term test is a precision propagation residual test.

Definition 3 (Minimum propagation residual test). *Given a signal propagating through a region for which curvature structure can be independently estimated, the first-order Dillon model predicts that residual timing or phase deviation, after standard relativistic, electromagnetic, environmental, and instrumental corrections, should correlate with at least one normalized curvature-variable term in Eq. (7).*

Operationally, the test requires: (i) a defined propagation path or interferometric geometry; (ii) independently estimated curvature or effective curvature-state variables; (iii) null-control paths or configurations; (iv) uncertainty budgets for clock, phase, thermal, material, electromagnetic, and geometric effects; and (v) a pre-registered regression or model-comparison protocol.

Falsifiability Criterion 5 (Minimum empirical failure condition). *If no statistically significant residual correlation is observed between propagation residuals and the normalized curvature-variable terms after standard corrections and null controls, then the first-order Dillon deviation model is falsified or strongly constrained for that experimental domain.*

This test is intentionally modest. It does not require validating the full Dillon Architecture. It only evaluates whether $\Delta c_{\text{eff}}/c$ contains measurable curvature-correlated structure beyond existing models and systematic errors.

18 Candidate Experimental and Observational Tests

Table 2: Falsifiability matrix for the first-order Dillon deviation model.

Domain	Observable	Dillon-model signature	Failure condition
Gravitational lensing	Lensing and time-delay residuals	Residuals correlate with curvature-gradient terms after mass-model uncertainties are controlled	No correlation beyond standard GR and lens modeling
Precision timing	Clock, pulse, or interferometric residuals	Timing offsets scale with normalized curvature or curvature-gradient structure	Residuals vanish or are explained by known systematic effects
Electromagnetic propagation	Phase or group-delay offsets	Structured-field or high-gradient environments produce repeatable phase residuals	Maxwellian propagation and apparatus effects fully explain observations
Quantum coherence	Phase coherence or decoherence shifts	Controlled geometric or acceleration conditions produce coefficient-bounded deviations	No reproducible shift after environmental controls
Cosmology	Redshift, expansion, or structure residuals	Large-scale residuals map to curvature-variable structure	Standard cosmology and astrophysical systematics explain residuals within uncertainty
Horizon physics	Near-horizon propagation signatures	Boundary behavior scales with curvature-evolution terms	No deviation beyond established semiclassical models

19 Comparison with Existing Frameworks

19.1 General Relativity

General relativity remains the required local and classical geometric limit. The Dillon Equation does not replace the Einstein field equations. Instead, it proposes an additional effective-propagation layer that must reduce to relativistic behavior when deviation terms vanish.

19.2 Quantum Field Theory

Quantum field theory remains the successful framework for particle interactions and field quantization. The present paper does not derive the Standard Model. Its relevance to quantum theory is limited to the hypothesis that quantum phase, coherence, and measurement behavior may exhibit effective propagation residuals under curvature-variable conditions.

19.3 String Theory and Quantum Gravity Programs

String theory, loop quantum gravity, causal set theory, asymptotic safety, and related programs attempt to reconcile quantum structure and gravity through distinct mathematical routes [11, 16, 17, 10]. The Dillon framework differs by beginning with an effective propagation observable rather than with quantized geometry, extended strings, or path-integral sums. It should therefore be judged not as a substitute for those programs, but as a proposed phenomenological bridge that could either constrain or motivate deeper models.

20 Limitations and Open Problems

The present framework has several limitations:

1. The compact function f is underdetermined without a domain-specific model.
2. The interpretation of K must be made precise in each physical setting.
3. Coupling coefficients are not known a priori and require empirical estimation.
4. The relationship between curvature-variable propagation and quantum measurement remains provisional.
5. The framework must be tested against existing high-precision constraints on Lorentz invariance, equivalence-principle behavior, gravitational lensing, and electromagnetic propagation.

These limitations are not incidental. They define the research program required to turn the Dillon Equation from a framework into a mature physical theory.

21 Research Program

A rigorous development path should proceed in four stages.

Stage 1. Mathematical specification. Define K , R , ∇K , and $\partial_t K$ for a selected manifold and metric class.

- Stage 2. Coefficient bounding.** Use existing observations to bound $\alpha, \beta, \gamma, \delta$.
- Stage 3. Targeted experiment design.** Identify high-gradient or time-varying curvature environments where residuals may be most detectable.
- Stage 4. Cross-domain transfer.** Test whether coefficients or functional constraints inferred in one domain predict behavior in another.

22 Independent Reproducibility Materials

A mature submission should be accompanied by reproducibility materials. At minimum, these materials should contain symbolic derivations, simulation notebooks, parameter sweeps, synthetic test cases, negative-control datasets, configuration files, versioned analysis scripts, and a description of how uncertainty budgets are propagated. Where phase-coupled or cyclic observables are modeled, the materials should also include toroidal/geodesic simulation notebooks, ablations comparing Euclidean and compact-manifold embeddings, reachability-corridor maps, and null controls in which the 4D predictive-geometry layer is disabled. Where quantum-measurement or electromagnetic-propagation examples are used, the materials should include mode-basis assumptions, link or measurement boundary definitions, coherence-scoring rules, and replayable proof bundles. Where lunar-descent, energy, aerospace, or TEVR examples are used, raw traces, calibration records, null controls, instrument certificates, uncertainty budgets, and analysis hashes should be archived with enough metadata to permit one-command replay of the analysis. This requirement is not an accessory to the theory; it is part of the falsifiability standard.

23 Conclusion

The Dillon Equation is proposed as a curvature-variable propagation framework in which local causal invariance is preserved while effective propagation across structured physical regions may be modeled as a function of curvature state, geometric curvature, curvature-gradient structure, and temporal curvature evolution. The central observable is not a local variation of c , but the effective deviation

$$\Delta c_{\text{eff}} = c_{\text{eff}} - c.$$

The framework is scientifically meaningful only if it satisfies three conditions: it must recover known physics as a limiting case, it must identify measurable deviations, and it must be falsifiable. The normalized first-order model introduced here provides an initial path toward empirical constraint:

$$\frac{\Delta c_{\text{eff}}}{c} \approx \alpha \frac{K}{K_0} + \beta \frac{R}{R_0} + \gamma \frac{|\nabla K|}{G_0} + \delta \frac{|\partial_t K|}{T_0}.$$

If future measurements constrain all such coefficients to zero, the first-order model fails. If nonzero curvature-correlated residuals are detected and replicated across domains, the Dillon Equation would provide a candidate bridge between geometric gravitation, effective field propagation, quantum-informational behavior, and cosmological structure. At present, it should be regarded as a formal, testable research program rather than a completed final theory.

A Plain-Language Summary

The framework preserves Einstein locally. It does not say that a local observer measures a different speed of light. It says that when propagation is inferred across extended, curved, changing, or

structured geometry, there may be an effective propagation difference. That difference is Δc_{eff} . The scientific question is whether this deviation is real, measurable, and correlated with curvature variables after known physics and systematic errors are accounted for.

B Compact Equation Set

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K), \quad (8)$$

$$\Delta c_{\text{eff}} = c_{\text{eff}} - c, \quad (9)$$

$$c_{\text{eff}} = c F\left(\frac{K}{K_0}, \frac{R}{R_0}, \frac{|\nabla K|}{G_0}, \frac{|\partial_t K|}{T_0}\right), \quad (10)$$

$$\frac{\Delta c_{\text{eff}}}{c} \approx \alpha \frac{K}{K_0} + \beta \frac{R}{R_0} + \gamma \frac{|\nabla K|}{G_0} + \delta \frac{|\partial_t K|}{T_0}. \quad (11)$$

C Mathematical Closure Work and Public Review Archive

The related mathematical corpus is included as an archive map rather than as a load-bearing premise. Its role is to document the author’s public proof and validation pathway and to clarify where terminology such as closure, leak, rigidity, attractor, admissibility, residual family, and bounded endgame originates.

Table 3: Omega-Genesis and Omega Calculus materials relevant to public review.

Material	Role in the broader Dillon Architecture	Role in this journal paper
Omega-Genesis public repository [32]	Public archive for manuscripts, code, simulations, and validation artifacts related to Omega-Genesis and Dillon Architecture proof programs.	Public review and reproducibility context only.
Omega Calculus compact companion [30]	Formal language layer for Omega states, flow, closure, leaks, rigidity response, and attractor convergence.	Terminology and methodology context.
P versus NP proof manuscript [31]	Author-claimed structural proof using information manifolds, modular Ricci flow, residual families, and finite endgame closure.	Companion mathematical-closure reference; not a premise of the physics claim.
From Collatz to Cure [28]	Broad narrative of the Omega-Genesis, structural proof, cryptographic defense, and self-healing systems architecture.	Architecture lineage and claim-boundary reference.

Material	Role in the broader Dillon Architecture	Role in this journal paper
RH Bottom-Core Endgame [29]	Conditional and target-theorem manuscript using compression, classification, bounded core, and closure grammar.	Supplemental example of reduction grammar only.

D Related Dillon Framework Manuscripts

This appendix lists related Dillon Framework manuscripts by citation role. The purpose is to document the originating research program without making the present paper depend on every claim in the broader framework.

Table 4: Related Dillon Framework manuscripts and citation roles.

Manuscript	Relevance to this paper	Recommended citation role
Dillon Architecture full manuscript stack [21]	Originating program for CVP, the Dillon Equation, and cross-domain curvature-state grammar.	Main-body context only; not proof of the present propagation model.
CVP-Higgs manuscript [22]	Related curvature-response application to mass stabilization while preserving the Standard Model baseline.	Main-body related work for response-operator lineage.
4D Electromagnetic Curvature platform [23]	Applied curvature-state, scheduling, safety, and evidence-sealing grammar.	Main-body application context and future validation environment.
4D Innovation white paper [24]	Toroidal embedding, harmonic control planes, geodesic modulation, targeted outcome alignment, validation tiers, and ablation structure.	Main-body methodological context for simulation and reproducibility.
Predictive Geometry manuscript [25]	Reachability corridors, instability corridors, attractor guidance, and early structural warning.	Main-body methodological context for forward-state modeling.
Predictive Quantum Measurement Overlay [26]	Quantum-memory-enhanced measurement, 4D adaptive EM propagation control, canonical c_{eff} standard, and measurement-corridor framing.	Main-body source for measurement-roadmap and propagation-control context.

Manuscript	Relevance to this paper	Recommended citation role
CVP Artemis lunar descent [33]	Narrow applied test domain using the same c_{eff} expression and $C(K)$ operator under mascon-rich curvature gradients.	Main-body candidate validation domain; not proof.
From Collatz to Cure [28]	Broad architecture narrative, canonical equation family, and explicit caution that broad claims require independent review and replication.	Main-body architecture-caution reference.
Omega Calculus compact companion [30]	Symbolic language layer for closure, leaks, rigidity response, attractor convergence, and structural determinism.	Main-body/appendix methodology context; not a physical proof premise.
Omega-Genesis public repository [32]	Public archive for manuscripts, simulations, validations, and code related to the Omega-Genesis proof corpus.	Reproducibility and peer-review archive reference.
P versus NP proof manuscript [31]	Author-claimed proof manuscript using information manifolds, modular Ricci flow, residual families D/S/R/C, and finite endgame closure.	Supplemental proof-corpus reference; not load-bearing for the Dillon Equation.
Geodesic Shield [27]	TEVR evidence replay, deterministic classification, governance boundary, and human-authority-bounded decision intelligence.	Supplemental verification and governance reference.
RH Bottom-Core Endgame [29]	Reduction grammar example: compression, classification, bounded core, and closure.	Supplemental methodological analogy; not physics support.
AeroHarvest PRD [34]	Applied multi-source energy-harvesting demonstrator using 4D scheduling and AeroTEVR verification.	Supplemental engineering roadmap.
Dillon Cryptography [35]	Curvature-native security, BCDE, TEVR, and post-quantum architecture visuals.	Supplemental architectural context unless a text-based cryptography manuscript is prepared.

Manuscript	Relevance to this paper	Recommended citation role
Tesla Resonance to Dillon Curvature [36]	Readability rewrite integrating canonical notation, energy-platform flow, 4D scheduling, and TEVR claim discipline.	Supplemental energy-platform and notation context.

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